

Adapting Foundation Models for Astronomical Time Series Classification: Setting 1 on the LSST/PLAsTiCC Benchmark

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Abstract

We address 14-class astronomical object classification on the LSST/PLAsTiCC benchmark (The PLAsTiCC Team, 2018) a multivariate time series dataset of shape $(N, T=36, C=6)$ with severe class imbalance (155:1 ratio) and sparse observations. Following Setting 1, we adapt two pre-trained foundation models: MOMENT-1-large (Goswami et al., 2024) and Chronos-T5-Small (Ansari et al., 2024). We further compare with UniTS (Gao et al., 2024), a non-pretrained dual-attention architecture trained with data augmentation. Our best ensemble combining Chronos, InceptionTime $\times 5$, PatchTST, and MultiROCKET via F1-weighted soft voting achieves **64.48% accuracy** and **Weighted F1 = 0.61**, surpassing the published MUSE SOTA of 63.6% (Ruiz et al., 2021). Code: <https://github.com/coulzie031/DeepLearningForTimeSeries>

1. Introduction

The LSST (Large Synoptic Survey Telescope) dataset, derived from the PLAsTiCC challenge, is one of the most challenging multivariate time series classification (TSC) benchmarks. Each light curve records the brightness of an astronomical object across 36 epochs and 6 photometric filters (passbands u, g, r, i, z, y), yielding a tensor of shape $(N, 36, 6)$. Two factors make this dataset particularly hard: (i) *sparse observations* leading to NaN values, and (ii) *extreme class imbalance* class 53 (kilonovae) has only 7 training samples out of 2,459 total.

Foundation models pre-trained on large time-series corpora offer a natural approach to scarce-data regimes. We explore two such models under *Setting 1*: adapt a pre-trained founda-

tion model to the classification task. We further investigate UniTS (Gao et al., 2024), a from-scratch dual-attention architecture combined with targeted data augmentation, as a strong competitor. Finally, we combine complementary models through an ensemble to push accuracy beyond the published SOTA.

2. Dataset and Preprocessing

LSST/PLAsTiCC. The dataset contains 2,459 training samples, 2,466 test samples, and 14 classes (supernovae subtypes, variable stars, AGN, kilonovae, microlensing events). Labels are heavily imbalanced: the dominant class (SN Ia, class 90) represents 44% of training data while class 53 has 7 samples.

Preprocessing. Missing values are filled via forward-fill, backward-fill, then zero-fill. A binary mask (B, T, C) encodes observation positions. All channels are z-normalized using training statistics. To address imbalance, we use a `WeightedRandomSampler` with inverse-frequency weights and Focal Loss ($\gamma = 2$) (Lin et al., 2017), which downweights easy majority examples.

Data augmentation (applied only during training): *Jitter* adds Gaussian noise ($\sigma=0.05$) with 50% probability; *scaling* multiplies each channel by a random factor in $[0.9, 1.1]$; *time shift* applies a circular roll of ± 3 steps with 30% probability; *MixUp* ($\alpha=0.3$) interpolates pairs of samples and their labels.

3. Methods

3.1. Baseline: InceptionTime

InceptionTime (Ismail Fawaz et al., 2020) applies multi-scale 1D convolutions to each channel independently. We use kernel sizes (9, 19, 39), 2 residual Inception blocks, $F=32$ filters, and Global Average Pooling totalling 474k parameters. No pre-training is used. Trained with Focal Loss, AdamW, and cosine annealing, early-stopped on validation macro F1.

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3.2. Setting 1 Foundation Model Adaptation

MOMENT-1-large (Goswami et al., 2024). A 341M-parameter T5-based encoder pre-trained on over 1B timesteps from diverse domains. We pad sequences to $T=40$ (multiple of patch size 8), extract per-channel embeddings of dimension 1024, and flatten to a 6144-dim vector fed to an MLP head. Training proceeds in two phases: (1) linear probing backbone frozen, only the head trains (200 epochs, early stop ep62, val F1 = 0.303); (2) partial unfreeze last 4 of 24 T5 encoder blocks unfrozen with $LR=5 \times 10^{-6}$ (100 epochs, early stop ep26, val F1 = 0.303). Despite native support for classification, MOMENT fails to improve: 341M parameters is an ill-matched scale for 1,967 training samples.

Chronos-T5-Small (Ansari et al., 2024). A 46M-parameter T5-based model originally pre-trained for univariate probabilistic forecasting on real and synthetic time series. We discard the decoder and repurpose the encoder for classification analogous to using BERT for sentence classification. Each of the 6 passbands is tokenized independently; masked mean-pooling over hidden states yields a 512-dim per-channel embedding; the 6 embeddings are concatenated into a 3072-dim representation, which is fed to a 3-layer MLP head. Training: phase 1 linear probing (200 ep, stop ep86, val F1 = 0.433); phase 2 partial unfreeze of last 4 encoder blocks, $LR_{\text{head}}=5 \times 10^{-5}$, $LR_{\text{enc}}=5 \times 10^{-6}$ (100 ep, stop ep32, val F1 = 0.433). Chronos outperforms MOMENT decisively despite being designed for forecasting, confirming that model scale matters more than task alignment when data is scarce.

3.3. Competitor: UniTS with Data Augmentation

UniTS (Gao et al., 2024) is a NeurIPS 2024 model designed for unified multi-task time series learning. Its architecture interleaves three operations per block: (i) *sequence attention* over the time axis; (ii) *variable attention* across channels; (iii) a *Dynamic Linear Operator* (DLO) that performs low-rank temporal mixing via a factorized weight matrix $W=UV^T$, $U \in \mathbb{R}^{T \times r}$, $V \in \mathbb{R}^{T \times r}$. A learnable [TASK] token is prepended to the sequence. We use $d=128$, 8 heads, 6 blocks, DLO rank $r=16$ (~ 1.6 M params). Trained with label smoothing ($\varepsilon=0.1$), AdamW, cosine annealing, and the three augmentations described above. UniTS achieves the highest single-model accuracy (66.06%) and macro F1 (0.5314).

3.4. Additional Models

InceptionTime-Large $\times 5$. Five independently seeded instances of a larger InceptionTime ($F=64$, 3 blocks, 1.8M params), trained with distinct data splits and augmentation seeds. Test-time augmentation (TTA, $N=5$ passes) is ap-

plied. The averaged probabilities yield $\text{acc} = 0.6456$.

PatchTST (Nie et al., 2023). Channel-independent Transformer: each channel is split into non-overlapping patches of length 4 (9 patches for $T=36$), linearly projected to $d=64$, and processed by 3 Transformer encoder layers with 4 heads. Classification head pools over patch representations. Trained with MixUp and TTA. Achieves $\text{acc} = 0.5985$, macro F1 = 0.4599.

MultiROCKET (Dempster et al., 2020). 79,968 random convolutional kernel features followed by a RidgeClassifierCV. Non-neural, no augmentation. Achieves $\text{acc} = 0.6079$.

3.5. Ensemble: F1-Weighted Soft Voting

Each neural model outputs a probability vector $\mathbf{p}_i \in \mathbb{R}^{14}$ over classes. We combine models via weighted soft voting:

$$\hat{y} = \arg \max_c \sum_i w_i p_{i,c}, \quad w_i = \frac{\exp(10 \cdot F1_{\text{val},i})}{\sum_j \exp(10 \cdot F1_{\text{val},j})} \quad (1)$$

Models with validation macro F1 < 0.33 are excluded. The final ensemble includes InceptionTime-Large $\times 5$ ($w=0.852$), PatchTST ($w=0.068$), Chronos ($w=0.051$), and MultiROCKET ($w=0.029$). MOMENT is excluded (val F1 = 0.303).

4. Results

Table 1 summarises all results on the LSST test set (2,466 samples). Macro F1 is low due to two structurally unlearnable classes: class 53 (7 train samples, F1 = 0) and class 64 (24 samples, F1 = 0.08). Excluding these two classes, macro F1 rises to ≈ 0.49 . Weighted F1 = 0.61 (weighted by class support) better reflects real-world performance on the learnable 90% of test samples.

Foundation model comparison. Chronos (46M) significantly outperforms MOMENT (341M): +22.5 pts accuracy, +15.4 pts macro F1. The key factor is scale: at 1,967 training samples, a 341M-parameter model is prone to underfitting even under linear probing, while Chronos’ compact encoder fine-tunes effectively. Chronos also surpasses the baseline (macro F1 0.433 vs. 0.375), validating Setting 1.

UniTS vs ensemble. UniTS achieves the highest single-model scores ($\text{acc} = 0.6606$, $F1 = 0.5314$), notably outperforming our ensemble on macro F1. This gap stems from UniTS’ dual-attention mechanism explicitly modeling inter-channel dependencies, combined with targeted augmentation. However, the bake-off SOTA (Ruiz et al., 2021) reports only accuracy, making F1 comparisons with external baselines impossible. Our ensemble beats MUSE on accuracy (+0.88%) while UniTS further surpasses it (+2.46%).

Table 1. Test set results on LSST (14-class). Acc = accuracy, Mac.F1 = macro F1, W.F1 = weighted F1. SOTA results from Ruiz et al. (2021) report accuracy only.

Method	Acc	Mac.F1	W.F1
<i>Baselines (no pre-training)</i>			
InceptionTime	0.5483	0.3753	
InceptionTime + aug.	0.5783	0.4324	
<i>Foundation models (Setting 1)</i>			
MOMENT-1-large	0.3078	0.2791	
Chronos-T5-Small	0.5333	0.4333	
<i>Competitor models</i>			
PatchTST + aug. + TTA	0.5985	0.4599	
MultiROCKET	0.6079	0.3625	
UniTS + aug.	0.6606	0.5314	
InceptionTime-L $\times$ 5 + TTA	0.6427	0.4150	
Ensemble (ours)	0.6448	0.4200	0.61
<i>Published SOTA (Ruiz et al., 2021)</i>			
MUSE	0.636		
ROCKET	0.632		
MrSQL	0.603		

Class imbalance. Both approaches substantially benefit from imbalance handling. Standard InceptionTime from the bake-off (no imbalance handling) achieves only 34.0% (Ruiz et al., 2021); our baseline with WeightedSampler and Focal Loss reaches 54.8% a +20.8 pt gain from preprocessing alone.

5. Conclusion

We present a comprehensive study of foundation model adaptation (Setting 1) and complementary architectures on the LSST benchmark. Our key findings are: (1) Chronos-T5-Small, despite being designed for forecasting, transfers effectively to multivariate classification, outperforming MOMENT by a large margin confirming that model scale relative to training set size matters more than pre-training task alignment; (2) UniTS with data augmentation achieves the best single-model macro F1 (0.5314), driven by its dual sequence-variable attention and low-rank temporal mixing; (3) F1-weighted soft voting over diverse models surpasses MUSE SOTA on accuracy; (4) imbalance handling (WeightedSampler + Focal Loss) is the single most impactful preprocessing decision on this dataset.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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